FDS EXPERIMENT-12

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Experiment to detect outliers in given dataset.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
data = {
    "Marks" : np.array([45,50,52,48,49,51,47,53.50,48,52,49,150])
    }

df = pd.DataFrame(data)
print(df)
```

```
    Marks
0    45.0
1    50.0
2    52.0
3    48.0
4    49.0
5    51.0
6    47.0
7    53.5
8    48.0
9    52.0
10   49.0
11  150.0
```

```python
Q1 = df["Marks"].quantile(0.25)
Q3 = df["Marks"].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[
    (df["Marks"] < lower_bound) |
    (df["Marks"] > upper_bound)
]
```

```python
print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)
print("Outliers:")
print(outliers)
```

```
Q1: 48.0
Q3: 52.0
IQR: 4.0
Lower Bound: 42.0
Upper Bound: 58.0
Outliers:
    Marks
11  150.0
```

```python
plt.boxplot(df["Marks"])
plt.title("Box Plot for Outlier Detection")
plt.ylabel("Marks")
plt.grid(True)
plt.show()
```

